

15.1 Exam Review

Four modules. Name the method, then the number that would have caught the failure.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

The exam is cumulative. It asks you to **name a method** and **say what you would measure**. Trivia without a metric is not the point of DATA 443/643.

Write the same story the lectures taught. Name the method, say why it exists, sketch the boxes, write the formula, and name a number that would have caught the failure. Fluency is not a check: RAG without recall@k, a GAN without coverage, and CoT without faithfulness are incomplete answers.

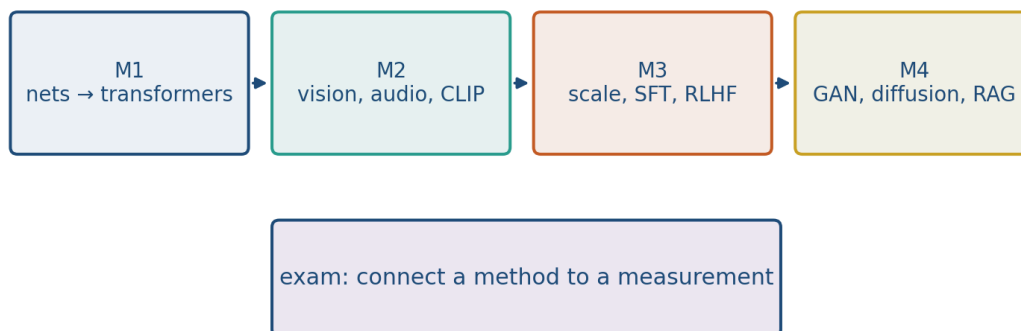
The sitting is a closed-book written exam plus the remaining talks. The formula sheet is what you can write, not a trivia list. Practice in that shape: “same face, low loss_G” wants mode bins; “cite [d5], fact in [d1]” wants attribution plus recall@k. Naming a paper without a metric scores poorly.

1. Four modules

Module	Weeks	Spine
1	1–3	Neurons, embeddings, RNNs/gates, attention, GPT vs BERT
2	4–6	Multimodal, ViT patches, contrastive / CLIP / BLIP, audio as tokens
3	7–10	Scaling and efficiency, SFT and adapters, RLHF / DPO, safety and editing
4	11–14	GANs and mode coverage, diffusion and latent text conditioning, CoT / vote / faithfulness, RAG and ReAct

You should be able to draw the picture (two GAN players; $q(x_t | x_0)$; retrieve-then-generate) and attach **one number** to it (coverage, a denoising loss, recall@k, majority-vote accuracy, a faithfulness rate).

Four modules, one stack. The project sits on top of all four.



Exam shape: name the method, then the number that would have caught it.

failure	measure
same face, low lossG	mode bins
right box, last step 52	faithful among wins
cite [d5], fact in [d1]	attribution + recall@k

2. How to study

Rework the labs on CPU: XOR and embeddings, long-range gates, one attention head, toy CLIP retrieval, 2-D GAN coverage, one reverse diffusion step, vote-and-flag traces, bag-of-words RAG. Redo each note's practice questions out loud. For every named method, write one sentence of the form "I would change X ; I would report Y on split Z ."

Formulas worth being able to write:

- neuron: $a = \sigma(w^\top x + b)$
- attention: $\text{softmax}(QK^\top / \sqrt{d})V$
- GAN: $\min_G \max_D \mathbb{E}[\log D(x)] + \mathbb{E}[\log(1 - D(G(z)))]$
- diffusion jump: $x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \varepsilon$
- CFG: $\hat{\varepsilon} = \varepsilon_u + s(\varepsilon_c - \varepsilon_u)$
- cosine: $a^\top b / (\|a\| \|b\|)$
- DPO-shaped check: $\sigma(0) = 1 / 2$ when the two log-ratio terms match

You do not need to memorize paper years. You do need to say **what you would plot** after you write the formula.

A compact board checklist:

Failure you see	Method that was in play	Number that would have caught it
All fakes look like one face	GAN	coverage / mode bins, not lossG
Pretty pictures, missed prompt	latent diffusion + CFG	CLIP image-text cosine and s
Right box, wrong algebra	CoT	faithfulness rate among correct boxes
Vote of a popular wrong number	self-consistency	shared-bug check; k vs accuracy plot
Cite [d5], fact only in [d1]	RAG	attribution + recall@k
Calc returned 408, answer 428	ReAct	observation used? parse log

3. What the exam asks

Typical item: given a failure (collapsed modes, ignored chunk, lucky CoT answer, biased CLIP hit), **name the method that was in play** and **the measurement** that would have caught it. Another item: pick GPT versus BERT, CLIP versus BLIP, GAN versus diffusion, CoT versus ReAct, and justify with the task plus the metric.

Lab → exam shortcuts: Lab 11 bins = coverage; Lab 12 oracle reverse step = you know x_0 ; Lab 13 last-integer vs box = faithfulness; Lab 14 cosine rank + `calc log` = RAG + ReAct. If you can redo those four on paper, Module 4 is in shape.

Project language belongs here: baseline, one justified change, ablation, failure case. That is the same spine as note **14.3**.

Remaining talks

Canvas is official for who speaks when. If you still owe a presentation, bring a **failure case** (input, output, what broke), not only a demo that worked. Upload what Canvas asks. Time limits on the hub still apply.

4. Teaching this note

This is a **review map**, not a new method. About **40 minutes** at the board walking the four-module table and the failure→metric grid, then **~15 minutes** of Karpathy as a recap of the LLM stack (not of GANs/diffusion). Leave 20 minutes for student questions and remaining talks. Weeks 13–14 are CoT, vote/search, faithfulness, RAG, and ReAct; Weeks 11–12 are GANs and diffusion.

Board order:

- **0–12 min.** Modules 1–2: neuron, embedding, attention, CLIP cosine. One picture each.
- **12–24 min.** Modules 3–4: SFT/RLHF/DPO in one sentence each; GAN vs diffusion; CoT vs RAG vs ReAct.
- **24–40 min.** Failure table. Worked exam-style item (GAN collapse; unfaithful CoT; RAG citation).

Then play the Karpathy minutes below. **GANs and diffusion are not in that talk**; restudy notes **11.1–12.3** and the assigned clips from those weeks.

Bring one page of handwritten formulas and one page of “failure → metric” rows. That beats rereading every HTML slide the night before.

5. Worked example (one exam-style item)

Prompt: “Fakes are sharp. Every fake is the same face. Generator loss is low. What do you measure?”

Answer in the form the exam wants:

- **Method:** GAN (Week 11). Implicit sampler; D can be fooled by one mode.
- **Not a measurement:** loss_G (it can fall because D collapsed).
- **Measurement:** coverage — fraction of fakes nearest each real mode (Lab 11), or precision/recall in a feature space. Report both bins, not a mean.

A second item: "CoT accuracy 90%; a calculator disagrees with half the traces."

- **Method:** chain-of-thought; the bug is **unfaithful** traces (Week 13; Lab 13 already scored it).
- **Measurement:** among correct boxes, fraction whose parsed steps match the box. Self-consistency may still vote the lucky number.

A third item: "Top-2 cosine missed the gold chunk; the answer still cites it."

- **Method:** RAG. Retrieval recall@2 is 0 on that item.
- **Check:** attribution against the stuffed context, not against the whole corpus. Citing an id that was never retrieved is a fail.

Keep the answers short. The exam is not a blog post.

6. Where students get stuck

- Memorizing paper titles and then blanking on the metric.
 - Mixing CoT / self-consistency / ToT / ReAct as one name.
 - Using Karpathy as a substitute for Weeks 11–12. That video does not teach $q(x_t | x_0)$.
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7. Video

Watch [Andrej Karpathy, Intro to Large Language Models](#) as a **recap of the stack**. Map weeks to minutes; do not try to play the whole hour in class.

Minutes	What he is doing	Course weeks
0:20–11:22	Inference, then training on internet text	1–3 (net, tokens, next-token)
11:22–17:52	How they work; finetune into an assistant	1–3 architecture; 8 SFT
21:05–27:43	Comparisons, labels, RLHF; then scaling laws	7 scaling; 9 RLHF / DPO
27:43–33:32	Tool use: browser, calculator, interpreter	14 RAG (browser as retrieve) and ReAct
33:32–35:00	Multimodality	4–6 CLIP / vision / audio
35:00–38:02	Thinking, System 1/2	13 CoT / extra tokens
40:45–42:15	Custom GPTs, files	14 RAG as a private index
45:43–59:23	Security (high level)	10 red-teaming <i>as evaluation</i> , not as a cookbook

In class, play **11:22–17:52** and **27:43–38:02** (~17 min) if you only have one pass. Students who missed Week 7–9 should add **21:05–27:43**. Students who missed Week 13–14 should add **35:00–42:15**.

Same URL as notes **13.1**, **14.1**, and **14.2**. Deep Dive chapters from **13.2–13.3** are optional extra, not a replacement for the formula sheet.

Week 11–12 video recap (not Karpathy): generative models from **46:45** ([setup / train / collapse](#)); Umar Jamil diffusion [1sPXkm2NH4](#) from **7:35**; Stable Diffusion [ZBKpAp_6TGI](#) **0:00–45:00**. Talk craft: Peyton Jones [sT_-owjKlbA](#) **0:00–20:00**.

8. Practice

1. A GAN's fakes look sharp and every one is the same face. What do you measure besides looking?
2. A RAG answer cites [d5] but the claim only appears in [d1]. Method name, and the check.
3. CoT accuracy is high; a calculator disagrees with half the traces. What exam idea is that, and which week's lab already scored it?
4. $x_0 = 1$, $\bar{\alpha}_t = 0.64$, $\varepsilon = 2$. Write x_t . Which week is this, and is it a GAN step?
5. Three CoT answers 408, 238, 408. Majority? If all three traces compute $17 \times 20 = 340$ and still box those numbers, what does the vote fail to catch?